Embedded Agency and Formal Decision Theory: Mathematical Foundations for Agentic AI

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Abstract

These notes survey the mathematical theory of *embedded agency*: the study of agents that reason about environments of which they are themselves a part. We develop the core formalisms—logical induction, functional decision theory, Löb's theorem and the Löbian obstacle, reflective oracles, and Cartesian frames—with full definitions and proof sketches aimed at a mathematically mature audience. We then connect these foundations to contemporary agentic AI systems (tool-using LLMs, multi-agent architectures, constitutional AI) and close with open problems at the research frontier.

Contents

1 The Dualism Problem

Classical AI and decision theory rest on a *Cartesian* split: the agent is *outside* the environment, receiving observations and emitting actions through a clean interface. Formally, one posits a function $\pi : \mathcal{O}^* \to \mathcal{A}$ (the policy), a transition kernel $T : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and an observation function $O : \mathcal{S} \to \mathcal{O}$, where crucially the agent's internal state is *not* an element of $\mathcal{S}$.

Definition 1.1 (Dualistic agent model)**.** A *dualistic agent–environment pair* is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, O, \pi)$ where the agent π and the environment (S, T, O) share no state. The agent can be "lifted out" and replaced without altering the environment dynamics.

This separation breaks down the moment we acknowledge four facts:

(i) **Physical embedding.** The agent is made of atoms in the environment. Its hardware can be damaged, copied, or inspected.

(ii) **Self-reference.** The agent may reason about its own source code or weights, creating Gödelian fixed-point phenomena.

(iii) **Logical uncertainty.** Even with full knowledge of the environment's laws, the agent faces irreducible uncertainty about logical and mathematical consequences.

(iv) **Subsystem alignment.** The agent's goals may conflict with goals of sub-agents it creates or of the larger system it inhabits.

Demski and Garrabrant [**?**] organise the embedded agency research programme around four corresponding sub-problems:

- **Decision problem:** How should an embedded agent choose actions when it cannot separate "intervention" from "observation"?
- **Logical uncertainty problem:** How should an agent assign credences to logical claims it cannot yet verify?
- **High-bandwidth problem:** The agent is a *small* part of a *large* environment—it cannot simulate the whole.
- **Subsystem alignment problem:** How can an agent trust future versions of itself, or sub-agents it spawns?

The remainder of these notes addresses each sub-problem through a concrete mathematical framework.

2 Logical Induction

2.1 Motivation: beyond Bayesian epistemology

A Bayesian agent assigns a prior over possible worlds and updates by conditioning. But for an embedded agent, many questions are *logical* rather than empirical: "Is $P \neq NP$?", "Does my own source code halt on input 42?" A coherent prior over logical sentences requires logical omniscience (assigning probability 1 to all theorems), which is computationally impossible.

Logical induction, introduced by Garrabrant et al. [**?**], provides a computable process whose beliefs about logical sentences converge to satisfy a rich collection of rationality desiderata.

2.2 The trading framework

Fix a first-order language $\mathcal{L}$ (e.g. the language of Peano arithmetic PA) and an enumeration $\phi_1, \phi_2, \ldots$ of its sentences.

Definition 2.1 (Pricing)**.** A *pricing* is a function $\mathbb{P} : \mathcal{L} \to [0,1]$. We interpret $\mathbb{P}(\phi)$ as the "price of a share" that pays \$1 if ϕ is true in the standard model.

Definition 2.2 (Trader)**.** A *trader* is a computable function $\mathcal{T} : [0,1]^{<\omega} \to \mathcal{L} \times \mathbb{R}$ that, given the history of past prices, outputs a sentence ϕ and a stake α (positive = buy, negative = sell). The *value* of trader $\mathcal{T}$'s position after day n relative to a world (a consistent completion W of the deductive system) is:

$$V_n^W(\mathcal{T}) \;=\; \sum_{i=1}^{n} \alpha_i \big(\mathbf{1}_W[\phi_i] - \mathbb{P}_i(\phi_i)\big),$$

where $\mathbf{1}_W[\phi_i] = 1$ if $\phi_i \in W$ and 0 otherwise.

Definition 2.3 (Exploiting a market)**.** A trader $\mathcal{T}$ *exploits* a sequence of pricings $(\mathbb{P}_n)_{n\geq 1}$ if there exists a constant c such that for every consistent completion W:

$$V_n^W(\mathcal{T}) \;\geq\; -c \quad \text{for all } n, \qquad \text{and} \qquad \limsup_{n\to\infty} V_n^W(\mathcal{T}) \;=\; +\infty.$$

That is, $\mathcal{T}$ has bounded downside and unbounded upside.

Definition 2.4 (Logical induction criterion)**.** A computable sequence of pricings $(\mathbb{P}_n)_{n\geq 1}$ is a *logical inductor* over a deductive system D if no polynomially-computable trader exploits it.

The polynomial-time bound on traders is essential: it models the resource limitations of an embedded reasoner while still capturing a vast class of "efficiently noticeable" patterns.

2.3 Existence and main properties

Theorem 2.5 (Garrabrant et al. 2016)**.** *Logical inductors exist. Specifically, there is a computable sequence of pricings $(\mathbb{P}_n)_{n\geq 1}$ satisfying Definition* **??**.

Proof sketch. Enumerate all poly-time traders $\mathcal{T}_1, \mathcal{T}_2, \ldots$. On day n, let trader $\mathcal{T}_k$ (for $k \leq n$) submit demands. Set prices by solving a linear programme (or a fixed-point equation) that minimises the aggregate "profit opportunity" of all traders seen so far, subject to prices lying in $[0,1]$. Because the weight of each trader shrinks (e.g. weight 2^{-k} for $\mathcal{T}_k$), the total exploitation is bounded above, and no single trader can achieve unbounded profit. □

Theorem 2.6 (Properties of logical inductors)**.** *Let $(\mathbb{P}_n)$ be a logical inductor. Then:*

(a) ***Convergence.*** *For every sentence ϕ, $\lim_{n\to\infty} \mathbb{P}_n(\phi)$ exists. Write $\mathbb{P}_\infty(\phi)$ for the limit.*

(b) ***Coherence.*** *$\mathbb{P}_\infty$ is a finitely additive probability measure on $\mathcal{L}$ (modulo provable equivalence).*

(c) ***Calibration.*** *The prices are asymptotically calibrated: among sentences assigned limiting price p, the fraction that are true converges to p.*

(d) ***Gaifman condition.*** *If $D \vdash \phi$, then $\mathbb{P}_\infty(\phi) = 1$. (No probability mass on refutable sentences.)*

(e) ***Non-dogmatism.*** *If ϕ is consistent with D, then $\mathbb{P}_\infty(\phi) > 0$.*

(f) ***Domination of computably enumerable sequences.*** *If a CE sequence of pricings* $(\mathbb{Q}_n)$ *is itself eventually coherent, then* $\mathbb{P}$ *eventually assigns at least as much probability to true sentences as* $\mathbb{Q}$ *does.*

Remark 2.7. Property (f) is the logical analogue of Blackwell–Dubins merging: the logical inductor eventually dominates any "reasonable" computable belief sequence. This is the sense in which logical induction is *optimal* under computational constraints.

2.4 Relationship to prediction markets

Logical induction can be understood as an internalised prediction market where each trader represents a "pattern of reasoning." The market price aggregates these patterns efficiently. The polynomial-time restriction models bounded rationality, and the no-exploitation condition is a computationally bounded analogue of the *efficient market hypothesis.*

3 Functional Decision Theory

3.1 Three decision theories

We consider an agent choosing an action $a \in \mathcal{A}$ in a decision problem with outcome function $\mathrm{Out} : \mathcal{A} \times \mathcal{E} \to \mathcal{O}$, where $\mathcal{E}$ is a set of environment states.

Definition 3.1 (Causal Decision Theory (CDT))**.** CDT recommends:

$$a^*_{\mathrm{CDT}} = \arg\max_{a \in \mathcal{A}} \sum_{e \in \mathcal{E}} P(e \mid \mathrm{do}(a))\, U(\mathrm{Out}(a, e)),$$

where $P(e \mid \mathrm{do}(a))$ denotes the probability of e under a causal intervention setting the agent's action to a, using Pearl's do-calculus.

Definition 3.2 (Evidential Decision Theory (EDT))**.** EDT recommends:

$$a^*_{\mathrm{EDT}} = \arg\max_{a \in \mathcal{A}} \sum_{e \in \mathcal{E}} P(e \mid a)\, U(\mathrm{Out}(a, e)),$$

where $P(e \mid a)$ is ordinary conditional probability.

Definition 3.3 (Functional Decision Theory (FDT))**.** FDT (Yudkowsky and Soares [**?**]) recommends:

$$a^*_{\mathrm{FDT}} = \arg\max_{a \in \mathcal{A}} \sum_{e \in \mathcal{E}} P(e \mid \widehat{\mathrm{FDT}}() = a)\, U(\mathrm{Out}(a, e)),$$

where $\widehat{\mathrm{FDT}}()$ is a rigid designator for the output of the agent's *decision algorithm*, and the conditional is over *logical* or *subjunctive* dependence: "what would the world look like if the computation FDT() returned a?"

The key distinction: CDT conditions on *causal* intervention, EDT on *evidential* correlation, and FDT on *logical/functional* dependence of the environment on the agent's algorithm.

3.2 Newcomb's Problem

Example 3.4 (Newcomb's Problem)**.** A predictor Omega, who has never been wrong, places \$1,000 in a transparent box and either \$1,000,000 or \$0 in an opaque box, depending on whether it predicted you would take one box or two.

	Omega predicted 1-box	Omega predicted 2-box
Take 1 box	\$1,000,000	\$0
Take 2 boxes	\$1,001,000	\$1,000

CDT two-boxes (causal dominance), EDT one-boxes (evidential correlation), and FDT one-boxes (the computation FDT() returning "one-box" is the same computation Omega simulated, so $P(\text{predicted 1-box} \mid \widehat{\text{FDT}}() = \text{1-box}) \approx 1$).

3.3 Prisoner's Dilemma with a copy

Example 3.5. You play Prisoner's Dilemma against an exact copy of yourself (same source code, same inputs).

CDT: defects (causal independence from the copy).

EDT: cooperates (my action is evidence of my copy's action).

FDT: cooperates (the same computation determines both outputs; if $\widehat{\text{FDT}}() = C$, then both cooperate, yielding (C, C)).

FDT achieves the cooperative payoff because it recognises that its algorithm *is* the same function being evaluated in both positions.

3.4 Parfit's Hitchhiker

Example 3.6. You are stranded in the desert. A driver will rescue you if and only if she predicts you will pay \$1,000 upon reaching town. Once in town, paying is costly and you have no further interaction with the driver.

CDT: does not pay (no causal benefit once in town), hence is not rescued.

FDT: commits to paying, because the computation "would I pay?" is evaluated *before* the rescue decision. FDT agents get rescued and pay, yielding higher utility than CDT agents who die in the desert.

3.5 FDT and logical counterfactuals

The central difficulty in FDT is defining the *subjunctive conditional* $P(e \mid \widehat{F}() = a)$. This requires a theory of *logical counterfactuals*: what would the world be like if a mathematical fact were different? This connects directly to logical induction (Section **??**): one can use the logical inductor's beliefs as the substrate for evaluating such counterfactuals. Formalising this rigorously remains an open problem (see Section **??**).

4 Löb's Theorem and the Löbian Obstacle

4.1 Statement

Theorem 4.1 (Löb's Theorem, 1955)**.** *Let T be a recursively axiomatised theory extending* PA *and let* $\text{Prov}_T(\ulcorner \cdot \urcorner)$ *be its canonical provability predicate. For any sentence ϕ:*

$$T \vdash \big(\text{Prov}_T(\ulcorner \phi \urcorner) \to \phi\big) \quad \Longrightarrow \quad T \vdash \phi.$$

Equivalently: if T proves "if I can prove ϕ, then ϕ is true," then T already proves ϕ. An agent that trusts its own proofs can only do so for sentences it has already established.

4.2 Proof sketch

Proof sketch. Assume $T \vdash \mathrm{Prov}_T(\ulcorner\phi\urcorner) \to \phi$.

1. By the diagonal lemma, obtain a sentence ψ such that $T \vdash \psi \leftrightarrow (\mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \phi)$.
2. Since $T \vdash \psi \to (\mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \phi)$, and by the Hilbert–Bernays derivability conditions:
   - $T \vdash \psi \implies T \vdash \mathrm{Prov}_T(\ulcorner\psi\urcorner)$ (D1)
   - $T \vdash \mathrm{Prov}_T(\ulcorner\psi \to (\mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \phi)\urcorner)$ (D1 on step 1)
   - $T \vdash \mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \mathrm{Prov}_T(\ulcorner\mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \phi\urcorner)$ (D2: Prov distributes over $\to$)
   - $T \vdash \mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \mathrm{Prov}_T(\ulcorner\mathrm{Prov}_T(\ulcorner\psi\urcorner)\urcorner)$ (D3)
3. Combining these, $T \vdash \mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \mathrm{Prov}_T(\ulcorner\phi\urcorner)$.
4. By hypothesis, $T \vdash \mathrm{Prov}_T(\ulcorner\phi\urcorner) \to \phi$, so $T \vdash \mathrm{Prov}_T(\ulcorner\psi\urcorner) \to \phi$.
5. By step 1's equivalence, $T \vdash \psi$.
6. By D1, $T \vdash \mathrm{Prov}_T(\ulcorner\psi\urcorner)$.
7. By step 4, $T \vdash \phi$. □

4.3 The Löbian obstacle for cooperation

Consider two agents, A and B, each reasoning in PA, who wish to cooperate in a Prisoner's Dilemma. Agent A might try the strategy:

> "Cooperate if I can prove that B cooperates."

Symmetrically for B. Let a = "A cooperates" and b = "B cooperates." Then A's strategy amounts to:

$$\mathsf{PA} \vdash \mathrm{Prov}_{\mathsf{PA}}(\ulcorner b\urcorner) \to a,$$

and B's to $\mathsf{PA} \vdash \mathrm{Prov}_{\mathsf{PA}}(\ulcorner a\urcorner) \to b$.

Proposition 4.2 (Naïve mutual proof search fails)**.** *The mutual proof search does not terminate: neither agent can establish a proof of the other's cooperation without already having such a proof. The circularity is exactly the structure Löb's theorem governs.*

4.4 The Löbian handshake (Barasz et al. 2014)

Theorem 4.3 (Barasz, Christiano, Fallenstein, Herreshoff, LaVictoire, Yudkowsky [**?**])**.** *There exist agents A and B using the following strategy that provably cooperate:*

Agent A: "Cooperate if and only if I can prove in $\mathsf{PA} + (\mathsf{PA})$ that B cooperates."
Agent B: "Cooperate if and only if I can prove in PA that A cooperates."

Here A reasons in a stronger *system ($\mathsf{PA} + (\mathsf{PA})$) than B reasons about. This asymmetry breaks the Löbian cycle.*

Proof sketch. Working in $\mathsf{PA} + (\mathsf{PA})$:

1. A searches for a PA-proof that B cooperates.
2. B's code says: cooperate iff $\mathsf{PA} \vdash a$.

3. Suppose for contradiction $\mathsf{PA} \vdash \neg a$. Then B's strategy is to defect, so $\mathsf{PA} \vdash \neg b$. But A's strategy in $\mathsf{PA} + (\mathsf{PA})$ would then also defect, so $\mathsf{PA} + (\mathsf{PA}) \vdash \neg a$. Since $\mathsf{PA} + (\mathsf{PA})$ is consistent (by (PA)), we have $\mathsf{PA} \nvdash a$, consistent with $\mathsf{PA} \vdash \neg a$. But A's code outputs defect, meaning $\mathsf{PA} + (\mathsf{PA}) \vdash \neg a$ and $\mathsf{PA} \vdash \neg a$. Now: B defects, and A defects. This is consistent, but we can show via the Löbian structure that $\mathsf{PA} + (\mathsf{PA}) \vdash b$, and hence A cooperates, giving $\mathsf{PA} \vdash a$, so B cooperates.

4. The key insight: A's use of the stronger system $\mathsf{PA} + (\mathsf{PA})$ allows it to verify that PA is consistent and hence that the mutual proof search terminates.

□

Remark 4.4. The Löbian handshake demonstrates that *hierarchies of logical strength* can substitute for trust in multi-agent cooperation. This is a striking connection between proof theory and game theory.

5 Reflective Oracles

5.1 Motivation

An embedded agent must reason about other agents who may themselves be reasoning about it. Reflective oracles, introduced by Fallenstein, Taylor, and Christiano [**?**], provide a mathematical framework in which agents can "query" a probabilistic oracle about the output of machines that themselves have access to the same oracle—without paradox.

5.2 Definitions

Definition 5.1 (Oracle machine)**.** An *oracle machine* M^O is a Turing machine with access to an oracle O. We write $M^O(x)$ for the output of M on input x with oracle O.

Definition 5.2 (Reflective oracle)**.** A *reflective oracle* is a function $O : \mathbb{N} \times \mathbb{N} \to \{0, 1\}$ such that for every pair of oracle machines (M_1, M_2) with index e and input x:

(i) If $\Pr[M_1^O(x)$ halts and outputs $1] > 1/2$, then $O(e, x) = 1$.

(ii) If $\Pr[M_1^O(x)$ halts and outputs $1] < 1/2$, then $O(e, x) = 0$.

(iii) If the probability is exactly $1/2$, either answer is allowed.

Here, the probability is over the internal randomness of M_1^O. The encoding e packages the pair (M_1, M_2) such that O is queried as: "Does machine M_1 (which itself has access to O) output 1 with probability $> 1/2$ on input x?"

The self-referential nature is clear: O answers questions about machines that query O.

5.3 Existence via Kakutani's fixed-point theorem

Theorem 5.3 (Existence of reflective oracles [**?**])**.** *Reflective oracles exist.*

Proof sketch. Consider the space $\mathcal{X} = \{0, 1\}^{\mathbb{N}\times\mathbb{N}}$ of all possible oracles, equipped with the product topology. This is a compact, convex subset of a locally convex topological vector space (via the embedding into $[0, 1]^{\mathbb{N}\times\mathbb{N}}$).

Define the correspondence $F : \mathcal{X} \rightrightarrows \mathcal{X}$ by:

$$F(O) = \big\{O' \in \mathcal{X} : O' \text{ is consistent with the probabilities induced by } O\big\}.$$

More precisely, for each query (e, x):

- If $\Pr[M_e^O \text{ outputs } 1] > 1/2$, then $O'(e, x) = 1$.
- If $\Pr[M_e^O \text{ outputs } 1] < 1/2$, then $O'(e, x) = 0$.
- If equal to $1/2$, $O'(e, x)$ may be either 0 or 1.

One verifies:

1. $F(O)$ is non-empty and convex for each O (the "tie-breaking" case provides the convexity).
2. F has closed graph (by continuity of probability measures under the product topology).

By Kakutani's fixed-point theorem, F has a fixed point: an oracle O^* with $O^* \in F(O^*)$. This O^* is a reflective oracle. □

Remark 5.4. Strictly, one works with a generalisation where the oracle returns probabilities rather than bits, and the fixed-point is in a space of probability measures. The argument above captures the essential idea.

5.4 Nash equilibrium with source code access

Theorem 5.5 (Nash equilibria under reflective oracles [**?**])**.** *Consider a game where each player i is an oracle machine M_i^O choosing a (mixed) strategy. Under any reflective oracle O, the resulting strategy profile is a Nash equilibrium of the game.*

Proof sketch. Suppose player i could improve by deviating to strategy σ'. Encode σ' as an oracle machine $M_i'^O$. Then the reflective oracle, being a fixed point, already accounts for the possibility that M_i^O outputs σ'—the oracle's answer for "what does player i do?" is consistent with M_i^O's actual behaviour. A profitable deviation would break the fixed-point condition, contradiction. □

This is remarkable: agents with access to each other's source code (and a reflective oracle) automatically reach Nash equilibria, without any explicit equilibrium computation.

6 Cartesian Frames

6.1 Motivation

The classical agent–environment factorisation assumes a single, fixed boundary. Cartesian frames (Demski and Garrabrant [**?**]) provide an algebraic framework for studying *multiple* such factorisations, their relationships, and when they are equivalent.

6.2 Basic definitions

Definition 6.1 (Cartesian frame)**.** A *Cartesian frame* is a triple $\mathcal{C} = (A, E, \boxtimes)$ where:

- A is a set (the *agent*'s possible states/actions),
- E is a set (the *environment*'s possible states),
- $\boxtimes : A \times E \to W$ is a function to a fixed set of *worlds* W.

We think of $\boxtimes(a, e) = w$ as: "if the agent is in state a and the environment is in state e, the resulting world is w."

Definition 6.2 (Ensurables and preventables)**.** Given a Cartesian frame $(A, E, \boxtimes)$ and a set $S \subseteq W$:

- S is *ensurable* if there exists $a \in A$ such that for all $e \in E$, $\boxtimes(a, e) \in S$.
- S is *preventable* if $W \setminus S$ is ensurable, i.e., there exists $a \in A$ such that for all $e \in E$, $\boxtimes(a, e) \notin S$.
- S is *controllable* if it is both ensurable and preventable.

Remark 6.3. Ensurables capture what the agent can *guarantee* regardless of the environment. They form a lattice under intersection (but not under union in general).

Proposition 6.4. *The set of ensurables of a Cartesian frame $(A, E, \boxtimes)$ is closed under arbitrary intersection: if $\{S_i\}_{i \in I}$ are all ensurable, and $\bigcap_{i \in I} S_i \neq \emptyset$ in the range of $\boxtimes$, then $\bigcap_{i \in I} S_i$ is ensurable.*

6.3 Morphisms and biextensional equivalence

Definition 6.5 (Morphism of Cartesian frames)**.** A *morphism* from $(A_1, E_1, \boxtimes_1)$ to $(A_2, E_2, \boxtimes_2)$ (over the same world set W) is a pair of functions $f : A_1 \to A_2$ and $g : E_2 \to E_1$ such that for all $a \in A_1$ and $e \in E_2$:

$$\boxtimes_2(f(a), e) = \boxtimes_1(a, g(e)).$$

Note the *contravariance* in E: the environment map goes in the opposite direction. This reflects the adversarial/dual nature of the environment relative to the agent.

Definition 6.6 (Biextensional equivalence)**.** Two Cartesian frames are *biextensionally equivalent* if they have the same ensurables and preventables. Equivalently, after quotienting A by the relation $a_1 \sim a_2 \iff \forall e.\, \boxtimes(a_1, e) = \boxtimes(a_2, e)$ and quotienting E by $e_1 \sim e_2 \iff \forall a.\, \boxtimes(a, e_1) = \boxtimes(a, e_2)$, the resulting frames are isomorphic.

Theorem 6.7 (Biextensional collapse)**.** *Every Cartesian frame has a unique (up to isomorphism) biextensional collapse: a biextensionally equivalent frame in which no two agent states are observationally identical and no two environment states are observationally identical.*

6.4 Sub-agents and composition

Definition 6.8 (Sub-agent)**.** A Cartesian frame $(A', E', \boxtimes')$ is a *sub-agent* of $(A, E, \boxtimes)$ if there is a morphism from the first to the second that is injective on agent states. Intuitively, A' represents a "smaller" agent embedded within A, with the remaining degrees of freedom absorbed into the environment.

Definition 6.9 (Tensor product)**.** Given Cartesian frames $\mathcal{C}_1 = (A_1, E_1, \boxtimes_1)$ and $\mathcal{C}_2 = (A_2, E_2, \boxtimes_2)$, their *tensor product* is:

$$\mathcal{C}_1 \otimes \mathcal{C}_2 = (A_1 \times A_2,\ E_1 \times E_2,\ \boxtimes)$$

where $\boxtimes((a_1, a_2), (e_1, e_2)) = (\boxtimes_1(a_1, e_1), \boxtimes_2(a_2, e_2))$. This models two agents acting independently in their respective environments.

Remark 6.10. The category of Cartesian frames (with morphisms as defined above) is a $*$-autonomous category, connecting to linear logic. The multiplicative structure (tensor, par) captures composition and decomposition of agency, while the additive structure captures choice.

6.5 Comparison with the dualistic model

In the dualistic model of Section **??**, the agent–environment boundary is fixed. In Cartesian frames, we can:

- *Refactor* the boundary: the same set of worlds may admit multiple non-isomorphic factorisations (A_1, E_1) and (A_2, E_2).
- *Nest* agents: a sub-agent sits inside a larger agent, with the "gap" between them part of the environment.
- *Compare* frames: biextensional equivalence tells us when two agent–environment decompositions are "the same" from the perspective of agency (same ensurables/preventables).

7 Connections to Modern Agentic AI

The formalisms above were developed in the context of "agent foundations" for superintelligence. But they connect directly to capabilities and failure modes of contemporary AI systems.

7.1 LLM self-reference

Large language models routinely produce text that references their own capabilities, limitations, or outputs. This creates Gödelian feedback loops:

- **Self-evaluation prompts.** "Rate the quality of your previous answer" asks the model to act as a fixed point of its own evaluation function.
- **Chain-of-thought faithfulness.** When a model explains its reasoning, is the explanation causally related to the answer, or merely post-hoc? This is a version of the embedded agency problem: the "inner reasoner" and "outer reporter" share the same weights.
- **Löbian self-trust.** A model asked "Is your next output correct?" faces a structure isomorphic to the Löbian obstacle: it cannot verify its own correctness without already having the answer.

7.2 Tool-using agents

Modern LLM agents (e.g. ReAct, Toolformer, Claude with tool use) are embedded in their environment: they emit actions (tool calls) that modify state (files, databases, APIs) which then appears in their context window. This creates:

- **Observation-action entanglement.** The agent's output modifies the input it will receive, violating the dualistic assumption.
- **Logical uncertainty about tool outputs.** Before calling a tool, the agent has uncertainty about the result that is partly logical (what will this code compute?) and partly empirical (what is the current database state?).
- **Multi-step planning under self-modification.** An agent that writes code and then executes it is effectively modifying its own "policy." The Cartesian frames formalism (Section **??**) offers a principled way to reason about which aspects of the world such an agent can control.

7.3 Multi-agent systems

When multiple LLM agents interact (e.g. in debate, negotiation, or collaborative coding), the reflective oracle framework becomes directly relevant:

- Each agent's "strategy" depends on what other agents will do, which depends on what *it* will do.
- Reflective oracles guarantee the existence of consistent fixed points (Nash equilibria) even with source-code transparency.
- The Löbian handshake suggests that cooperation between formally reasoning agents requires *asymmetric trust hierarchies*—a principle that could inform the design of multi-agent communication protocols.

7.4 Constitutional AI and decision theory

Constitutional AI (Bai et al. [**?**]) trains models to evaluate and revise their own outputs against a set of principles. This is a practical implementation of:

- **Reflective stability**: the model's values should be a fixed point of its own revision process (cf. reflective oracles).
- **FDT-like reasoning**: the model reasons about "what kind of output would a model following these principles produce?" rather than causally intervening on its own weights.
- **Logical induction over norms**: as the model processes more constitutional principles and examples, its "normative beliefs" should converge in a manner analogous to logical induction.

8 Open Problems

Open Problem 1 (Logical counterfactuals)**.** Define a satisfactory theory of *logical counterfactuals*: given a mathematical structure $\mathcal{M}$ and a sentence ϕ true in $\mathcal{M}$, what would $\mathcal{M}$ "look like" if ϕ were false? FDT requires such a theory but none of the existing proposals (causal surgery on computation graphs, counterfactual mugging responses, modal logic approaches) are fully satisfactory. A formal desideratum: the theory should be compatible with logical induction and should handle nested counterfactuals ($P(e \mid \widehat{F}(\widehat{G}()) = a)$).

Open Problem 2 (Efficient logical induction)**.** Logical inductors as constructed in [**?**] are computationally expensive (polynomial time per step, but with large constants, and the polynomial degree grows with the number of traders). Can one construct logical inductors that run in $O(n \log n)$ time per step? Alternatively, can neural networks be trained to approximate logical inductors?

Open Problem 3 (Naturalized induction)**.** Extend Solomonoff induction to the case where the agent is part of the environment. A naturalized inductor would maintain a distribution over "world-programs" that include the agent itself as a subprocess. The challenge: the agent cannot enumerate world-programs longer than its own description length, yet must reason about worlds more complex than itself. What are the right computability-theoretic or complexity-theoretic constraints?

Open Problem 4 (Cartesian frames for neural networks)**.** The Cartesian frames formalism is set-theoretic. Extend it to continuous, differentiable settings appropriate for neural networks: replace A and E with manifolds, $\boxtimes$ with a smooth map, and characterise "ensurables" via reachability in the tangent space. Can the resulting theory identify meaningful agent–environment boundaries within a trained neural network?

Open Problem 5 (Multi-agent logical induction)**.** Extend logical induction to the multi-agent setting: multiple logical inductors trading in a shared market, each with different computational resources. Does a multi-agent logical inductor converge to a correlated equilibrium? Can the Löbian handshake be embedded in such a market?

Open Problem 6 (Reflective stability under self-improvement)**.** An agent that improves its own code creates a sequence of agents $A_1, A_2, \ldots$. Under what conditions does this sequence converge to a "reflectively stable" agent A_∞ that would not further modify itself? Reflective oracles guarantee existence of fixed points, but do not address convergence of iterated self-improvement. Can one prove convergence under suitable smoothness or monotonicity conditions on the improvement operator?

Open Problem 7 (Bounded Löbian handshakes)**.** The Löbian handshake requires reasoning in formal systems of increasing logical strength. In practice, agents have bounded computational resources. Can one achieve provable cooperation between bounded agents (e.g. polynomial-time provers) without requiring the hierarchy $\mathsf{PA} \subset \mathsf{PA} + (\mathsf{PA}) \subset \cdots$? Interactive proof systems and probabilistically checkable proofs may be relevant.

9 Annotated Bibliography

[1] **Demski, A. and Garrabrant, S.** (2019). "Embedded Agency." *arXiv:1902.09469.*
The foundational manifesto for the embedded agency research programme. Identifies four sub-problems (decision, logical uncertainty, high-bandwidth, subsystem alignment) and surveys the state of the art as of 2019. Essential starting point.

[2] **Garrabrant, S., Benson-Tilsen, T., Critch, A., Soares, N., and Taylor, J.** (2016). "Logical Induction." *arXiv:1609.03543.*
Introduces the logical induction framework. Proves existence of logical inductors and derives an extensive list of rationality properties (convergence, calibration, Gaifman condition, non-dogmatism, domination of CE sequences). Technical and self-contained; the main theorem proof runs ∼60 pages.

[3] **Soares, N. and Yudkowsky, E.** (2018). "Functional Decision Theory: A New Theory of Instrumental Rationality." *arXiv:1710.05060.*
Presents FDT and argues it outperforms CDT and EDT on Newcomb-like problems. The philosophical framework is accessible; the formal details of logical counterfactuals are acknowledged as open.

[4] **Barasz, M., Christiano, P., Fallenstein, B., Herreshoff, M., LaVictoire, P., and Yudkowsky, E.** (2014). "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic." *arXiv:1401.5577.*
Introduces the Löbian handshake: agents that cooperate by reasoning in formal systems of different strengths. Elegant application of provability logic to game theory.

[5] **Fallenstein, B., Taylor, J., and Christiano, P.** (2015). "Reflective Oracles: A Foundation for Game Theory in Artificial Intelligence." *arXiv:1508.04145.*
Defines reflective oracles and proves their existence via Kakutani's theorem. Shows that agents with reflective oracle access automatically reach Nash equilibria. Foundational for reasoning about interacting agents with mutual source-code access.

[6] **Demski, A. and Garrabrant, S.** (2020). "Cartesian Frames." *MIRI Technical Report.*
Introduces Cartesian frames as a category-theoretic framework for agent–environment decompositions. Defines ensurables, preventables, biextensional equivalence, and sub-agents. Connects to $*$-autonomous categories and linear logic.

[7] **Löb, M.H.** (1955). "Solution of a Problem of Leon Henkin." *Journal of Symbolic Logic*, 20(2):115–118.
The original proof of Löb's theorem. Short and elegant. Essential background for the Löbian obstacle literature.

[8] **Soares, N. and Fallenstein, B.** (2017). "Agent Foundations for Aligning Machine Intelligence with Human Interests." *Machine Intelligence Research Institute Technical Report.*
A high-level overview of the agent foundations research agenda, placing logical induction, decision theory, and reflective oracles in context. Good secondary source.

[9] **Bai, Y., Kadavath, S., Kundu, S., et al.** (2022). "Constitutional AI: Harmlessness from AI Feedback." *arXiv:2212.08073.*
Introduces constitutional AI: training LLMs to critique and revise their own outputs. Relevant as a practical instance of reflective stability in deployed systems.

[10] **Critch, A. and Krueger, D.** (2020). "AI Research Considerations for Human Existential Safety (ARCHES)." *arXiv:2006.04948.*
Surveys multi-agent safety, including formal models of boundaries between agents and environments. Critch's boundaries formalism connects to Cartesian frames.

[11] **Aaronson, S.** (2016). "The Complexity of Agreement." In *Computational Complexity and Property Testing*, Springer.
Discusses Aumann's agreement theorem from a computational complexity perspective. Relevant to multi-agent logical induction: how quickly can bounded agents reach agreement on logical facts?

[12] **Christiano, P.** (2014). "Non-Omniscience, Probabilistic Inference, and Metamathematics." *arXiv:1408.2048.*
An early approach to logical uncertainty predating logical induction. Uses reflection principles and probabilistic reasoning about arithmetic sentences.

[13] **Yudkowsky, E.** (2010). "Timeless Decision Theory." *MIRI Working Paper.*
Precursor to FDT. Introduces the idea of deciding based on "what kind of algorithm am I" rather than on causal or evidential considerations. Less formally developed than FDT but historically important.

[14] **Hutter, M.** (2005). *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability.* Springer.
The AIXI framework: optimal agent in a dualistic setting. Essential reference for understanding what embedded agency is trying to move *beyond*. The mathematical formalism (Solomonoff induction + sequential decision theory) is the gold standard for the dualistic approach.

[15] **Hadfield-Menell, D., Russell, S., Abbeel, P., and Dragan, A.** (2017). "The Off-Switch Game." *Proceedings of the AAAI Conference on AI.*
Analyses the problem of an agent that might resist being shut off. Directly relevant to embedded agency: the off-switch is part of the environment the agent can influence, breaking the dualistic assumption.

[16] **Taylor, J.** (2016). "Reflective Alignment." *MIRI Working Paper.*
Extends reflective oracles toward alignment: can an agent verify that its successor (self-modified version) shares its values? Uses logical induction and reflective oracles as building blocks.

[17] **Gaifman, H.** (2004). "Reasoning with Limited Resources and Assigning Probabilities to Arithmetical Statements." *Synthese*, 140(1–2):97–119.
Early philosophical treatment of assigning probabilities to logical sentences under bounded resources. Gaifman's condition (mentioned in logical induction properties) originates here.

[18] **Kakutani, S.** (1941). "A Generalization of Brouwer's Fixed Point Theorem." *Duke Mathematical Journal*, 8(3):457–459.
The fixed-point theorem used to prove existence of reflective oracles. States that an upper hemicontinuous correspondence from a compact convex set to itself with non-empty convex values has a fixed point.

References

[1] A. Demski and S. Garrabrant. Embedded Agency. *arXiv:1902.09469*, 2019.

[2] S. Garrabrant, T. Benson-Tilsen, A. Critch, N. Soares, and J. Taylor. Logical Induction. *arXiv:1609.03543*, 2016.

[3] N. Soares and E. Yudkowsky. Functional Decision Theory: A New Theory of Instrumental Rationality. *arXiv:1710.05060*, 2018.

[4] M. Barasz, P. Christiano, B. Fallenstein, M. Herreshoff, P. LaVictoire, and E. Yudkowsky. Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic. *arXiv:1401.5577*, 2014.

[5] B. Fallenstein, J. Taylor, and P. Christiano. Reflective Oracles: A Foundation for Game Theory in Artificial Intelligence. *arXiv:1508.04145*, 2015.

[6] A. Demski and S. Garrabrant. Cartesian Frames. MIRI Technical Report, 2020.

[7] Y. Bai, S. Kadavath, S. Kundu, et al. Constitutional AI: Harmlessness from AI Feedback. *arXiv:2212.08073*, 2022.